higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MASONRY
NQF LEVEL 4

(12020524)

22 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only BLUE or BLACK ink.
6. Write neatly and legibly.

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QUESTION 1

QUESTION 1: GENERAL

Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 A … is used on construction projects to isolate or limit cracking caused by internal or external stresses on the structure.

A wall tie ⌘
B roof anchor
C super glue
D control joint

1.2 Freshly cured concrete is known as … concrete.

A abrasive
B green
C attach
D coat

1.3 A … is a masonry tool used to texturise stone and concrete.

A bush hammer
B rubber mallet
C wooden mallet
⌘ D chisel

1.4 Raking is the process of compacting mortar between the bricks to a depth of … mm.

A 100
B 205
C 10
D 75

1.5 The cement uses in plaster must comply with … 50197-1.

A (NCV)
B FET
C TVET
D SANS ⌘

1.6 … is a construction material used to fill the spaces between tiles and to provide support in tile joints.

A Plaster
B Screed
C Jointer
D Grout

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1.7 … is composed of two paper liners sandwiching a core, containing gypsum, starch, and some additives.

A Plaster-board
B Sand paper
C Cornice
D Skirting ⌘

1.8 ONE of the basic ingredients of grout is …

A soil.
B cement.
C stones.
D fine aggregate.

1.9 … are made from a mixture of clay and other natural materials.

A Glass tiles
B Ceramic tiles
C Galvanised tiles
⌘ D Corrugated tiles

1.10 Allow the grout to set for … hours before sealing the tiles.

A 2
B 72
C 24
D 3

(10 × 1) **[10]**

QUESTION 2: PLASTERING

2.1 FIGURE 1 below shows the surface of the concrete slab that has been prepared by applying a spatter-dash layer. ⌘



FIGURE 1

2.1.1 Why was spatter dash applied on the above concrete slab? (2)

2.1.2 How does one apply a spatter dash coat on concrete? (4)

2.1.3 Why is a bonding liquid added to spatter dash? (1)

2.1.4 If the substrate is too dry it needs to be pre-soaked.

Explain the method of pre-soaking the substrate before plastering. (7)

2.2 What is *off-shutter concrete*? (2)

2.3 What are the following plastering tools used for:

2.3.1 Wire brush

2.3.2 A straight edge

2.3.3 Block brush

2.3.4 Hand hawk

2.3.5 Plumb bob

(5 × 1) (5)

2.4 Define *wall chase.* (4)

2.5 Briefly describe the procedure used to mix a gauge of plaster mortar by hand. (5)

[30]

QUESTION 3: SCREED AND TOPPINGS

3.1 What is the datum level and how does it affect the level of the screed/topping? (4)

3.2 Describe the process followed to perform a slump test and use neat sketches to illustrate the test and the measure of a true slump. (10)

3.3 List THREE steps in preparing substrate to receive the screed or topping. (3)

3.4 How does one clean the substrate before laying screed? (2)

3.5 List FIVE steps in applying the slush coat. (5)

3.6 How does one remove grease from the substrate? (2)

3.7 Compaction has a structural as well as an attractiveness function.

Name FOUR types of compacting equipment that can be used on a large masonry site. (4)

[30]

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QUESTION 4: TILING

4.1 Discuss why waste outlets and drains should be blocked with rags to prevent ingress of tile adhesive. (6)

4.2 Name and explain TWO groups of testing used to test walls for soundness before applying tiles. (10)

4.3 Explain the importance of removing at least 80% of paint before the surfaces can be painted. (10)

4.4 Complete the following sentences by filling in the missing word or words. Write only the word or words next to the question number (4.4.1–4.4.4) in the ANSWER BOOK.

After the screed has been cast, the surface is covered with ... (4.4.1) to retain moisture for ... (4.4.2). After this period wait another ... (4.4.3) to ensure that the curing process is more or less complete before the surface is tiled. This means that ... (4.4.4) will pass between laying the screed or topping and tiling the floor surface. (4)

[30]

TOTAL: 100

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